Exp No: 14	Packet sniffing using RAW sockets

AIM:

To write a program using raw sockets to capture (sniff) the packets arriving at a network interface and to display the source IP address and the size of each packet.

ALGORITHM / PROCEDURE:

1. Find the IP address of the machine using `socket.gethostbyname(socket.gethostname())`.
2. Create a raw socket with `socket(AF_INET, SOCK_RAW, IPPROTO_IP)` and bind it to the IP address of the machine. Raw sockets need administrator rights.
3. Set the option `IP_HDRINCL` so that the IP header is included in the received packets.
4. Start an infinite loop and receive a packet with `recvfrom(65535)`.
5. Display the source IP address and the size of the packet in bytes.
6. Press Ctrl+C to stop the sniffer and close the socket.

DIAGRAM:

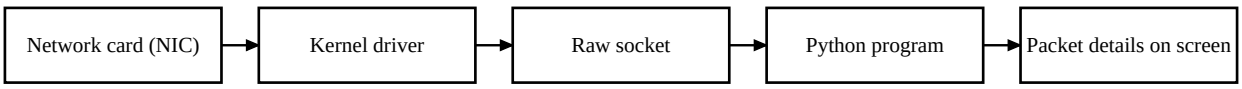


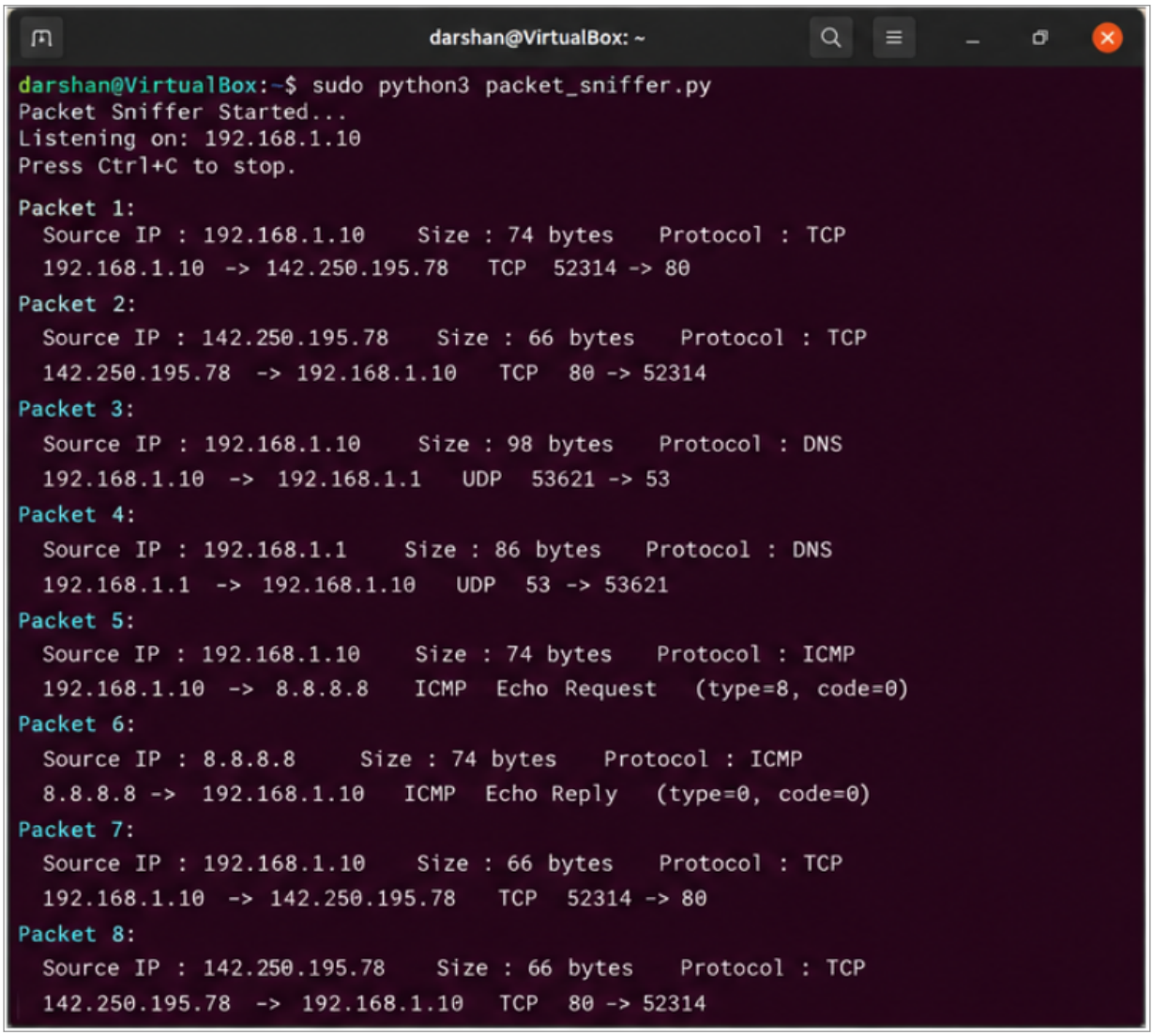
Fig 1: Flow of packet sniffing using a raw socket

PROGRAM:

packet_sniffer.py

```
import socket
HOST = socket.gethostbyname(socket.gethostname())
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
sniffer.bind((HOST, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
print("Packet Sniffer Started")
print("Listening on:", HOST)
print("Press Ctrl+C to stop")
try:
    while True:
        data, address = sniffer.recvfrom(65535)
        print("Source IP:", address[0])
        print("Packet Size:", len(data), "bytes")
        print("-" * 40)
except KeyboardInterrupt:
    print("\nPacket Sniffer Stopped")
finally:
    sniffer.close()
```

OUTPUT:



Terminal output of packet_sniffer.py showing the captured packets

RESULT:

Thus a packet sniffer was implemented using a raw socket. The source IP address and the size of the packets received by the interface were displayed.